The Global Coffee Health Dataset

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Explanation of Project: Why Coffee Matters?

Coffee is one of the world’s most consumed drinks, influencing daily habits, lifestyle, and health.

Health Effects

- **Benefits:** lower risk of heart disease, diabetes, some cancers; improved brain function.
- **Risks:** anxiety, poor sleep, stomach issues, high cholesterol, pregnancy concerns.

Research question

- Does a cup of coffee consumption have a statistically significant relationship with behavior and health (stress, sleep, BMI), and how can these patterns support decisions in public health and the coffee industry?

Explanation of Project: Why Coffee Matters? (Cont'd)

Dataset

- *Global Coffee Health Dataset (Kaggle), content* **10,000 records | 20 countries**
- Variables: coffee habits, stress, sleep, BMI, lifestyle

Methods

- Cleaned data
- Visualized patterns
- Ran correlation & significance tests
- Machine Learning: XGBoost Classification
- Utilized Classifiers to obtain feature related information

Key Outcome

- Coffee intake shows notable links with stress and health indicators.
- Results can guide health advice and industry decisions.

Analytical Techniques Used

Descriptive Statistics

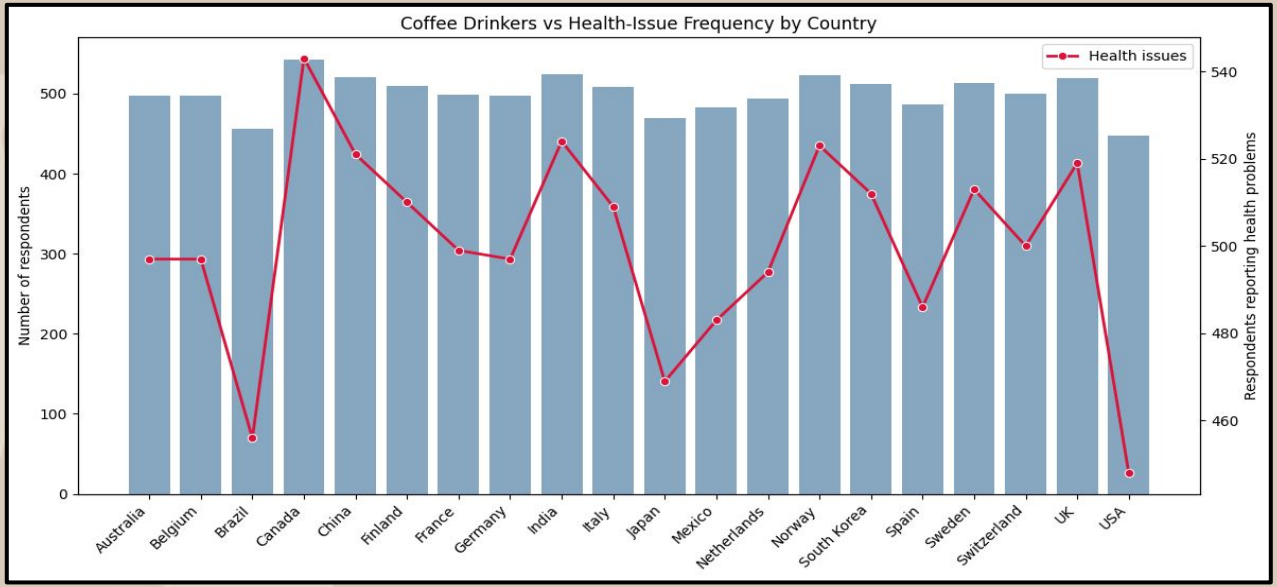
- Imputed all missing values with mode since missing values were all categorical
- Transformed categorical data into numeric data to be used in the correlation matrix and machine learning model
- Visualize distributions between variables in question using boxplots to show differences in mean, median, maximum, and minimum.
- Correlation matrix was created to obtain a better understanding of which variables had a higher chance of being statistically significant

Analytical Techniques Used (Cont'd)

Diagnostic Statistics

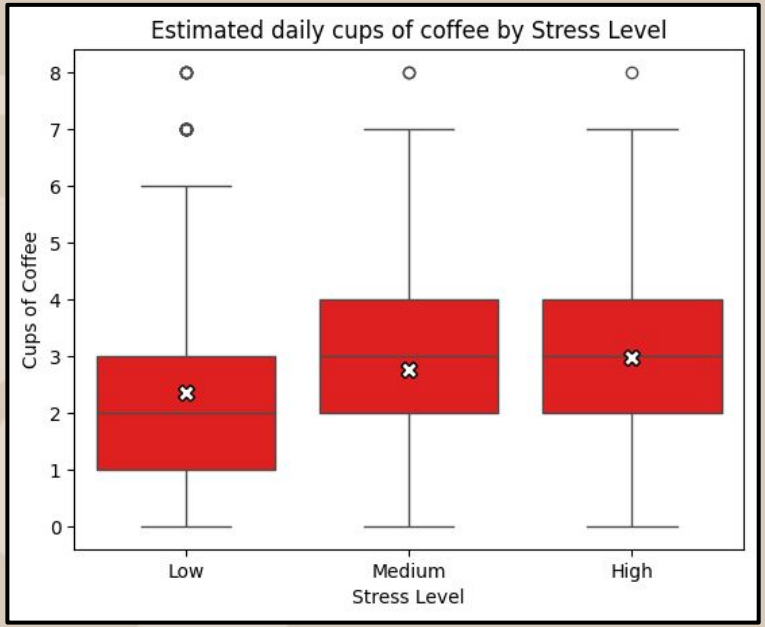
- Chi-Squared Test - tested for statistical significance
- Cramérs V - tested for effect size
- OLS Regression - tested for statistical significance once other variables are factored in
- XGBoost Classification - predicts which stress category a person belongs to from the given features
- Performed 5 fold cross validation to estimate the generalization error (0.979, 0.9795, 0.9795, 0.977, 0.98)), in addition to calculating mean squared error (0.022)

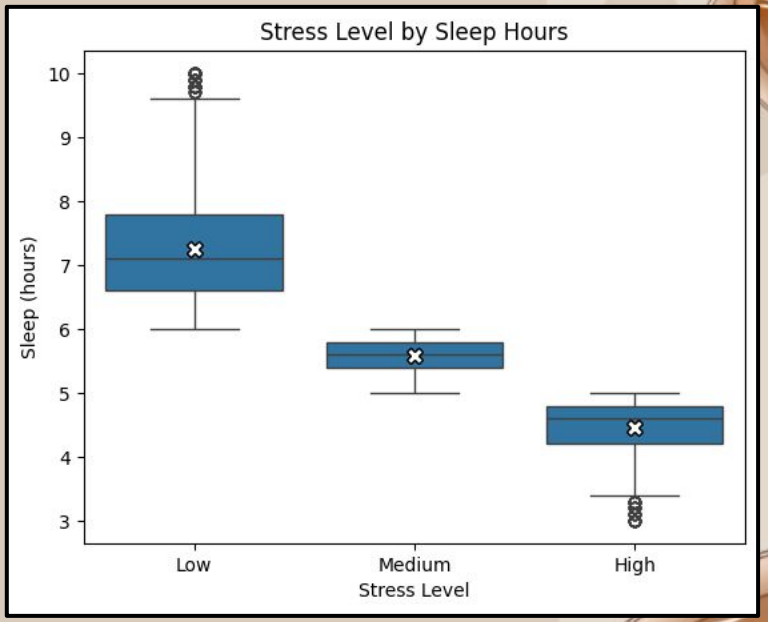
Visualization Techniques



This visualization explores how many countries in our dataset reported health issues. The bar chart shows how many respondents came from each country, while the line plot shows how many health-related problems were reported. This helped us see whether certain countries reported higher levels of health issues and did those countries also have a higher coffee intake?

Visualization Techniques





Boxplots helped us compare stress levels with two different factors. The first plot shows that coffee intake doesn’t change much across stress levels. The second plot shows that sleep hours drop noticeably as stress increases. This comparison helped us see that stress relates more strongly to sleep than to coffee consumption.

Major Findings: Do Cups of Coffee Predict Stress Levels?

- Coffee consumption and stress showed a statistically significant association confirmed through our chi-square test.
- However, the effect size was small (Cramer's V ≈ 0.11), meaning coffee alone does NOT strongly influence stress levels.
- OLS regression results showed cups of coffee was not a significant predictor of stress when controlling for sleep hours, sleep quality, health issues, and age ($p \approx 0.50$).
- Sleep-related variables were the strongest predictors of stress, especially sleep hours and sleep quality

Limitations

Our analysis of the Global Coffee Health Dataset was insightful, but there were several limitations that we encountered during our development.

- The synthetic nature of the dataset may limit real-world significance, since the patterns we observed don’t necessarily reflect actual human behavior.
- The ‘Health_Issues’ column had a whole lot of missing values, requiring imputation with the most frequent category, which could bias correlations involving health attributes.
- Converting categorical variables like stress level and sleep quality into numeric values simplified the data, but removed nuance.
- Feature selection was challenging, as including too many variables risked overfitting while excluding others weakened the model.
- The limited amount of resources prevented us from testing more advanced modeling techniques or using richer datasets.

Recommendations for Future Work

How to Strengthen This Analysis in the Future

- **Collect richer real-world data**: longitudinal tracking, biometric markers (cortisol, HRV, blood pressure), and more diverse demographics.
- **Use advanced modeling**: neural networks, ensemble methods, and deeper feature engineering to capture nonlinear relationships.
- **Improve validation**: extensive cross-validation to avoid overfitting and ensure stability.
- **Explore segmentation**: clustering to uncover lifestyle-based subgroups.
- **Move toward causation**: conduct controlled studies and include qualitative insights (self-reported stress, work conditions) for a holistic understanding.

Goal: Shift from correlation to actionable insights about behavior, stress, and physiological health.

Thank you!